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Stroke

Mike Bolshaw, Head of Claims

- Facts and figures
- What is a stroke?
- Risk factors
- Looking for a stroke
- Linking damage to deficit
- “Different strokes”
- The history of the definition
- Summary

“Stroke occurs approx.
152,000 times a year in the
UK....

“Men are at a 25% higher risk of
having a stroke at a younger age
compared to women...

“1 in 4 strokes in the UK occur in
people under 65 years old...

...once every 3 minutes and 27
seconds...”

...however, as women live longer
there are more total incidences
of stroke in women”

...the number of strokes in
those aged 20 to 64 increased
by 25% from 1990 to 2010
worldwide”



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What is a Stroke

Mike Bolshaw, Head of Claims

“A group of pathological conditions characterized by sudden, non-convulsive loss of neurological function due to brain ischaemia or intracranial haemorrhage”.

Stroke – resulting in permanent symptoms

Death of brain tissue due to inadequate blood supply or haemorrhage within the skull resulting in permanent neurological deficit with persisting clinical symptoms.

For the above definition, the following are not covered:

- Transient ischaemic attack.
- Traumatic injury to brain tissue or blood vessels.
- Death of tissue of the optic nerve or retina / eye stroke.

The mechanism of stroke

Vascular obliteration

A clot can be released from deposits and clog smaller or restricted vessels.

Bleeding

An artery or an aneurysm burst, the blood floods brain tissue.

Blood clot

Can form during atrial fibrillation, migrate to the brain and block vessels.

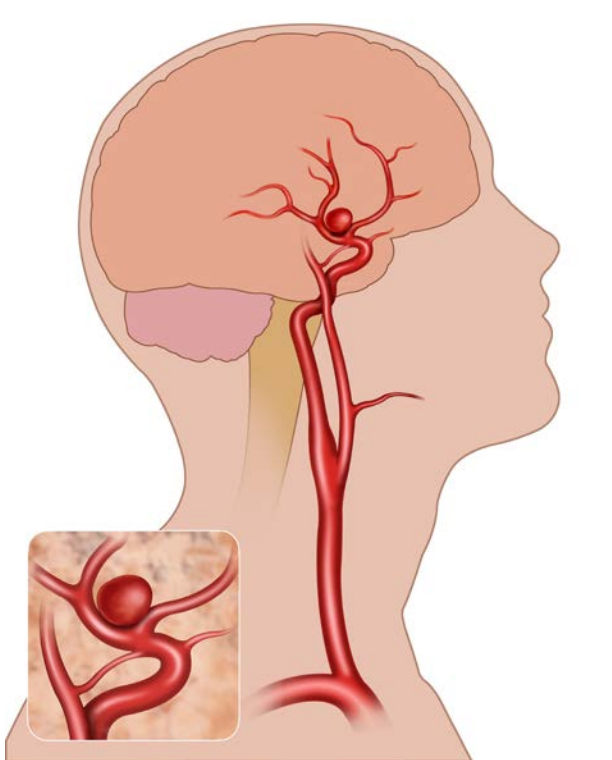


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- The brain is starved of oxygen and nutrients and brain cells die.
- The brain is placed under pressure by the bleeding and brain cells die.
- The damaged area of the brain leads to neurological symptoms ranging from significant and fatal to minor and temporary.
- Intervention is required urgently in order to prevent/minimise damage to the brain.
- Rehabilitation is required and may result in some return of function.

Public Health England's FAST campaign



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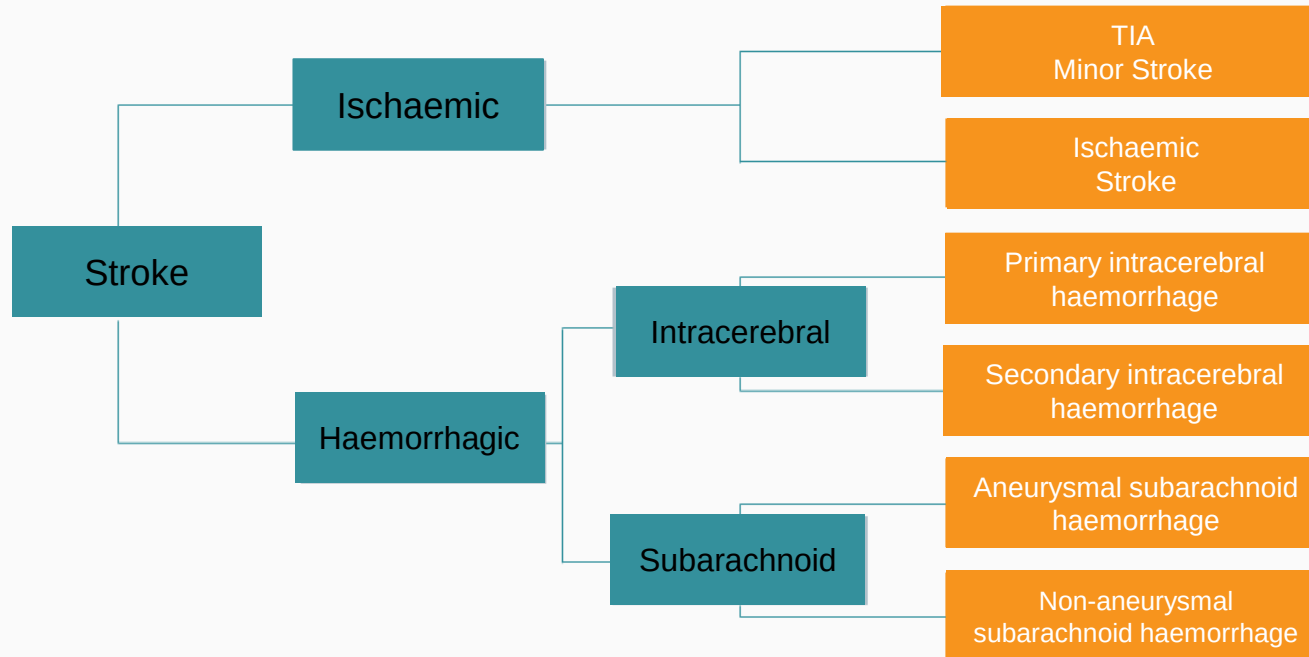
Facial weakness

Arm weakness

Speech problems

Time to call 999

Types of stroke



Types of stroke

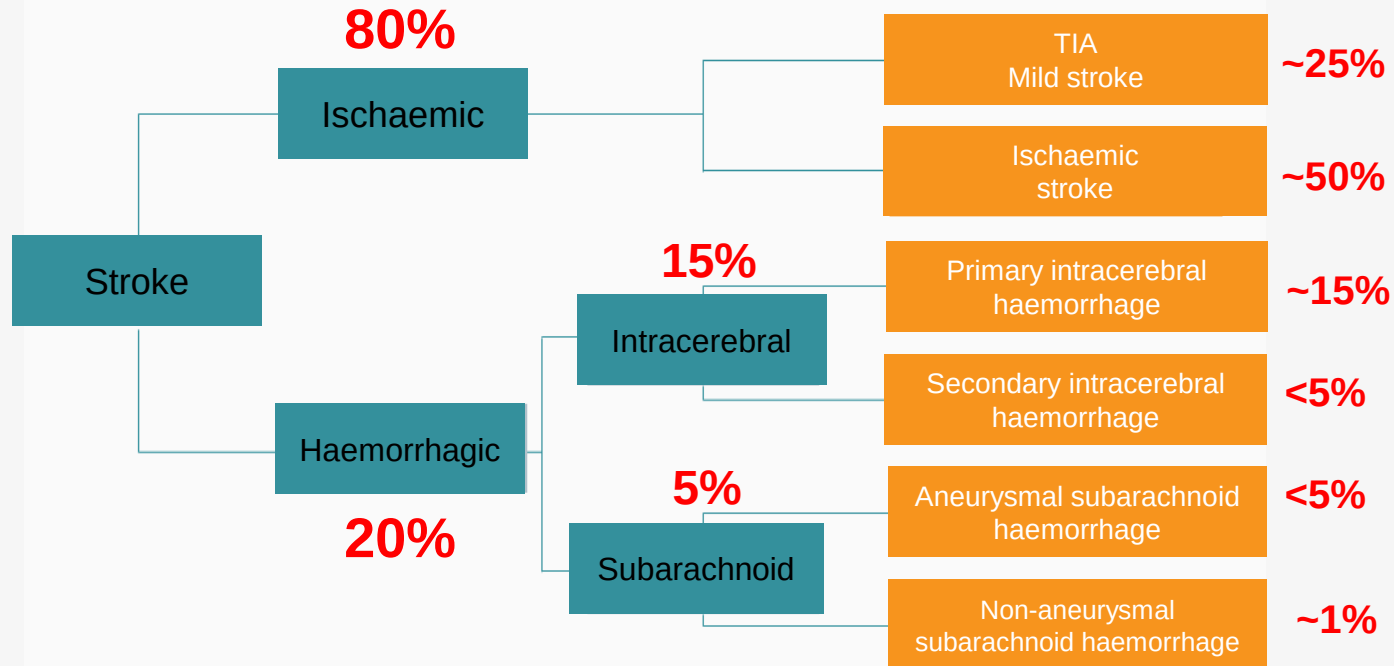




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Risk Factors

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Illicit drugs

Alcohol

Hypertension

Diabetes

Hypercholesterolaemia

Smoking

Oral contraceptive

Inactivity

However, there is also...

Age

Sickle cell disease

Ethnicity

Previous MI

Congestive Cardiac Failure

Family history

Cardiac arrhythmias – particularly
Atrial Fibrillation

Endocarditis



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Looking for a Stroke

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Has there been a bleed? Is there a blockage?
Has there been damage?

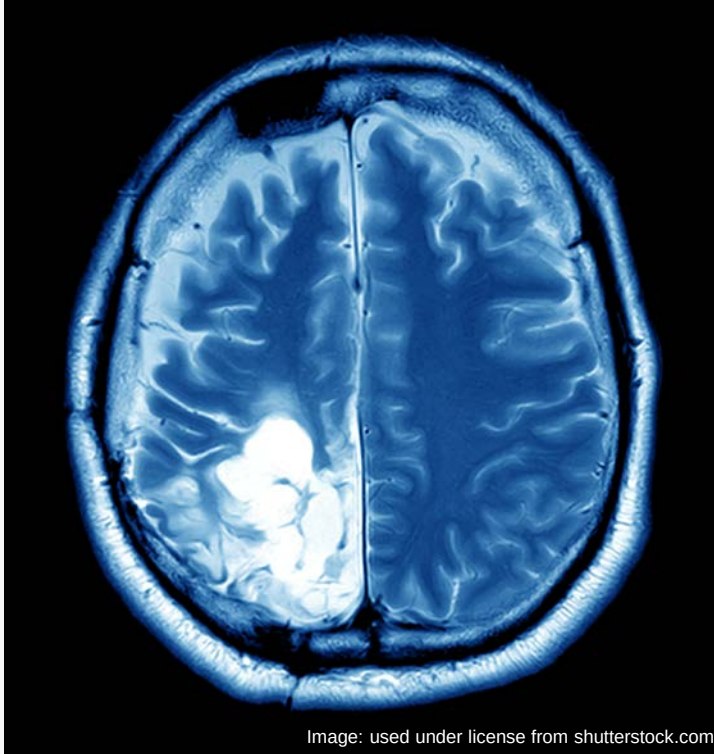
CT

MRI

CT angiography

Spinal tap

Carotid artery ultrasound

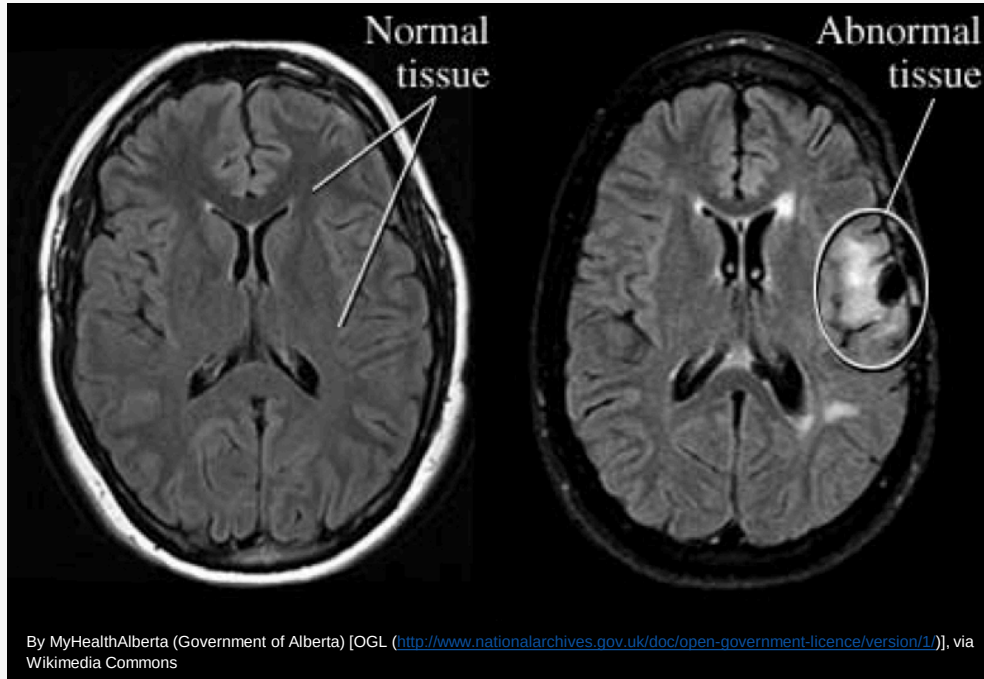


- CT scan of the brain (without contrast)
- Uses radiation (X-ray) to build a picture of the brain
- Fluid (CSF, blood) absorbs few X-rays so appears darker
- CT (with contrast) injection of dye to highlight blood flow more easily.



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- CT scan of the vascular supply (with contrast)
- Used to detect blockages or abnormalities that may cause ischaemia.



- Not typically used in the acute context
- Used to investigate what might be going on where unclear on CT or similar
- Can show areas of damage/death more clearly

Carotid artery duplex/doppler ultrasound



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- Not typically used in the acute context
- Used to observe arterial flow
- Nil risk/discomfort to patient

Spinal tap



- Sample of CSF extracted
- Analysed for red blood cells, xanthocromia (yellow colour), bilirubin via spectrophotometry
- NB: “bloody tap”



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Linking damage to deficit

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Know your lobes - volunteers please

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Lobes of the brain – location and function

Frontal

Reasoning, planning, parts of speech and movement (motor cortex), emotions, and problem-solving

Parietal

Perception of stimuli such as touch, pressure, temperature and pain, regulating the five senses, movement and visual orientation (R sided: ability to dress or groom oneself. L sided: language and perception)

Temporal

Perception and recognition of auditory stimuli (hearing) and long-term memory and processing new information (hippocampus)

Occipital

Many aspects of vision, movement and colour recognition, recognising objects, understanding language

In reality, the brain is even more complicated

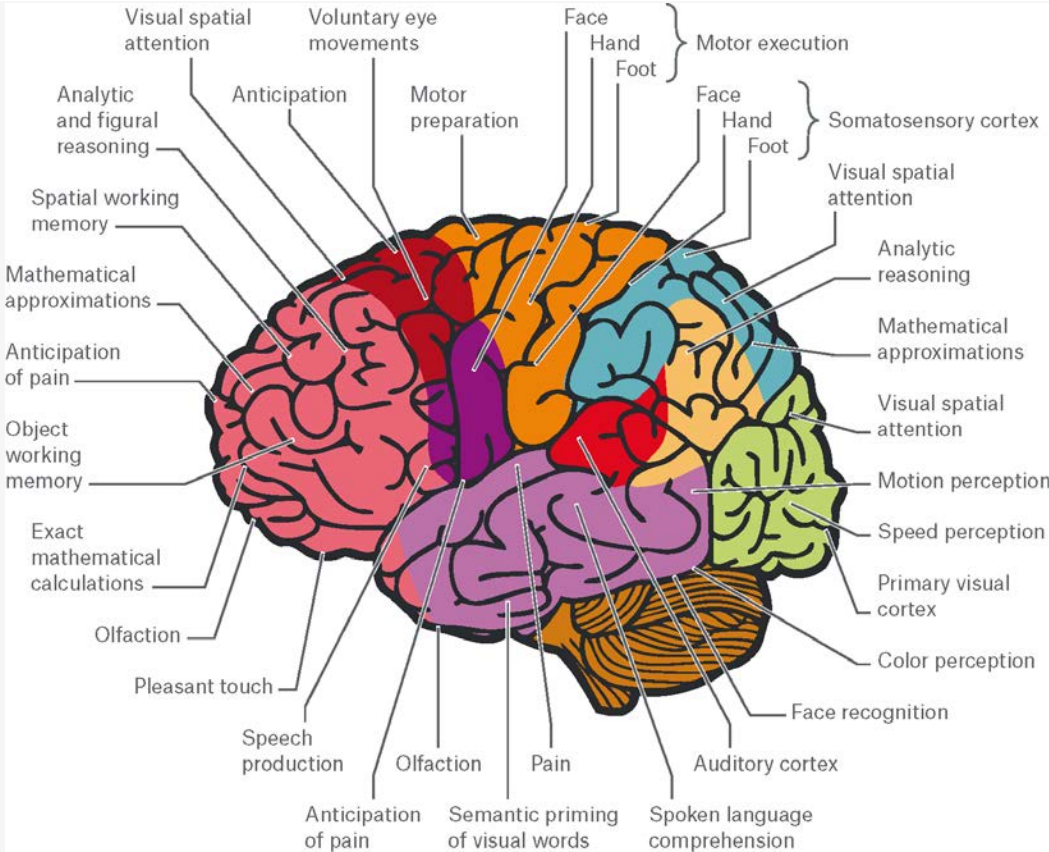




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“Different strokes”

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Subarachnoid haemorrhage

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Lacunar infarct

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Eye stroke and ear strokes

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Migrainous strokes

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The history of the definition

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Stroke – resulting in permanent symptoms

Death of brain tissue due to inadequate blood supply or haemorrhage within the skull resulting in permanent neurological deficit with persisting clinical symptoms.

For the above definition, the following are not covered:

- Transient ischaemic attack.
- Traumatic injury to brain tissue or blood vessels.
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However, some insurers now have wordings that don't look for permanent symptoms...

...they will look for symptoms lasting at least 24 hours.

History of the ABI definition

“....incident....
PN damage...”



“....incident....
PN damage...”



“....incident....
PN damage...”



“...**permanent symptoms**.
“...death of brain tissue...
within skull....
...PN deficit with persisting
clinical symptoms”



1999

2002

2004

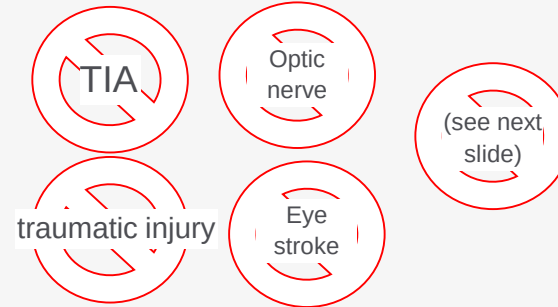
2006

“...**permanent symptoms.**
“...death of brain tissue...
within skull....
...PN deficit with persisting
clinical symptoms”



2011

“...**permanent symptoms.**
“...death of brain tissue...
within skull....
...PN deficit with persisting
clinical symptoms”



2014

Spot the difference (2006 and 2011 vs 2014)

Permanent neurological deficit with persisting clinical symptoms

~~Symptoms of~~ Dysfunction in the nervous system that are present on clinical examination and expected to last throughout the insured person's life.

~~Symptoms that are covered~~ To include numbness, hyperaesthesia (increased sensitivity), paralysis, localised weakness, dysarthria (difficulty with speech), aphasia (inability to speak), dysphagia (difficulty in swallowing), visual impairment, difficulty in walking, lack of coordination, tremor, seizures, ~~lethargy~~, dementia, delirium and coma.

The following are not covered:

- An abnormality seen on brain or other scans without definite related clinical symptoms
- Neurological signs occurring without symptomatic abnormality, e.g. brisk reflexes without other symptoms
- Symptoms of psychological or psychiatric origin.

- Stroke is common. Certainly in older ages, but it's becoming more common in younger lives.
- Know your risk factors: common or uncommon; to be disclosed or not disclosed; relevant to medical underwriting or not relevant to medical underwriting. Should you investigate misrep?
- Be aware of your own company's philosophy on more challenging 'stroke' scenarios and diagnoses: gather evidence/ask questions accordingly.
- Take into account *how* the diagnosis was made and *who* made it.
- Take into account who is reporting and assessing any PND with PCS. How does this relate to the original event?
- READ THE APPLICABLE DEFINITION, THEN READ IT AGAIN.



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Thank you

Mike Bolshaw, Head of Claims